Annals of Botany **81**: 233–250, 1998

A 3D Architectural and Process-based Model of Maize Development

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Received: 12 June 1997 Accepted: 24 September 1997

A 3D architectural and process-based model of maize development was implemented on the basis of the L-system software Graphtal, interfaced with physical models computing microclimate distributed on the 3D canopy structure. In a first step, we incorporated in the software Graphtal additional functions that enable bi-directional communication with external modules. A simple model for distributed photosynthetically active radiation and the model for apex temperature by Cellier *et al.* (*Agricultural and Forest Meteorology* **63**: 35–54, 1993) were interfaced with Graphtal. In a second step we developed a L-system model for maize, where production rules for growth and development of organs are based on the current state of knowledge of maize development as a function of temperature. Visual representation of the plant is based on the geometrical model of leaf shape by Prévot, Aries and Monestiez (*Agronomie* **11**: 491–503, 1991). Finally, various data sets were used to evaluate the physiological aspects and the geometrical representation. It is concluded that environmental L-systems are a convenient tool to integrate biophysical processes from organ to canopy level, and provide a framework to model growth of individual plants in relation to local conditions and ability to forage for resources. However, progress is needed to improve both the knowledge of physiological processes at the organ level and the calculation of physical environmental parameters; some directions for future research are proposed. © 1998 Annals of Botany Company

Key words: Growth model, 3D plant architecture, *Zea mays* L., corn, temperature, L-system modelling, developmental physiology, virtual plant.

INTRODUCTION

The characterization of biological and physical processes in canopy growth models is usually based on the description of the geometric structure as a continuous medium. This approximation enables the use of differential equations to describe mass and energy transfer between the plants and their environment. These models may give a detailed account of metabolism and growth in term of volumetric variables, however the physiological processes cannot be described at the level of individual organs or plants since only probabilistic descriptors such as volume density of leaf area or biomass are used.

In recent years, approaches have been developed to describe the geometric structure of plants in 3D. The development of 3D measurements in the field allows the reconstruction of geometric structure of actual crops (Moulia and Sinoquet, 1993; Ivanov *et al.*, 1995; Room, Hanan and Prusinkiewicz, 1996). On the other hand, models have been developed, based mainly on L-system or similar approaches, to simulate the 3D architecture of plants (Prusinkiewicz and Lindenmayer, 1990; Jaegger and de Reffye, 1992; Kurth, 1994). L-systems were initially designed to model the evolution of a set of cells (Lindenmayer, 1968). They progressively gained in versatility and were provided with graphical capability and now appear to be a powerful tool to model the growth of plants. They use 'production rules' acting on 'modules', to describe events occurring at a local level (organs and meristems for higher plants) and then the development of the whole organism resulting from these local processes. Production rules can be used to describe the qualitative changes involved in development, as well as the quantitative aspects of the growth of organs. At this level of analysis, a plant can be considered as a ramified structure, with 3D geometry but 1D topology. The connectivity relations can be accounted for in contextual L-systems, where the rules applied to a module may depend on the neighbouring modules. This enables communication from module to module. Contextual L-systems have been used to simulate transfer of information from an apex to buds and could, in principle, be applied to simulate transfer of mass or energy through the plant. Finally, one of the most appealing results of plant modelling through L-system is the ability to describe the apparent complexity of the whole plant as emerging from the parallel and recursive application of a small number of rules acting at the local level. Taking into account botanical knowledge to describe plant architecture and topology, and field measurements of geometrical parameters to describe size and orientation of organs, very realistic 3D representations of canopy geometry have been obtained.

The 3D geometry generated by L-systems can be used to describe light transfer and other aspects of microclimate as in the case of a structure reconstructed from field measurements. In principle, this enables computation of the microclimate distributed on individual organs, which can be used as an input for production rules featuring models of local physiological processes. However, until now, architectural plant models have mainly addressed the problem of the emergence of plant shape, and focused on the processes of ramifications, with a time step corresponding to the rhythm of production of new modules (de Reffye *et al.*,

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1988; Prusinkiewicz *et al*., 1997). In most cases, the size and number of leaves are defined to obtain a pleasant visual aspect with a tractable number of polygons, rather than to enable modelling of the biophysical interface between the plants and their atmospheric environment.

Some previous work addressed the problem of combining architectural and process based models. Guzy (1995) developed an architectural model for wheat, with growth driven by the model MODWht3 (Rickman, Waldam and Klepper, 1996). In this case, the architectural description was used as a visualization of MODWht3 simulations. Rapidel (1995) developed a model for transpiration using a 3D description of plant geometry, however plant shape was given *a priori*, and did not result from the modelled processes. Blaise (1991) considered the construction of the shape of trees interacting together using a probabilistic approach. Recent tree models (Perttunen *et al*., 1996; de Reffye *et al*., 1997) drive architectural development using yearly integrated variables (photosynthesis or transpiration). Mech and Prusinkiewicz (1996) presented examples of architectural plant models interacting with their environment, using simple approximations to describe biophysical processes.

This work is a further step to incorporate a mechanistic description of the bi-directional interactions between plants and their environment in architectural modelling. Such an approach may lead to a powerful tool to analyse crop growth, taking into account competition between organs at the plant level and competition between individual plants at the field level. We present here an architectural model for the growth of a population of maize plants, as a function of apex temperature. We used a time step of 3 h, compatible with a mechanistic description of both the physiological processes at the level of organs and the changes in plant environment. The temperatures of apices of individual plants are calculated at each time step by an energy-budget sub-model, using measured soil temperature, meteorological data and the 3D structure of the canopy generated at the previous time step.

The starting point to describe the growth processes was the temperature-driven module in CERES (Jones and Kiniry, 1986). However, in CERES, the structure is described statistically, in terms of leaf area expansion, and little attention is paid to the development of the stem. A specific feature of architectural models is the requirement to describe explicitly the start, the rate and the end of the growth of the different modules constituting the plant. Thus the parameterization of physiological processes in our model is based on a review of results published in the literature at the organ level. Data at the plant or canopy level were used for model evaluation.

Stem elongation of maize begins only after tassel initiation, so that in early growth stages, the apex is only a few centimetres above the soil and is strongly affected by soil temperature. Cellier *et al*. (1993) showed that, under clear sky conditions, soil temperature in a developing maize field could be as much as 20 °C higher than air temperature monitored by a standard meteorological station. Apex temperature was 5 to 10 °C warmer than air temperature monitored by a meteorological station. Thus using standard meteorological data or soil temperature to drive a growth model may strongly under- or overestimate the growth rate in the first stages of development. Here, apex temperature during early growth stage is calculated by an energy balance model (Cellier *et al*., 1993). We adapted the calculation of short-wave energy budget, to benefit from accurate computation of direct light, using the 3D geometric description of the plant. The model was validated on canopies with leaf area index (*lai*) below 0·5. On the other hand, when internodes begin to elongate, apex temperature approaches air temperature. Thus, in the present work, apex temperature is taken equal to air temperature after the first internodes have elongated.

Our model uses the L-system software Graphtal, which was modified to provide bi-directional communication with external models. The communication procedure and the model to compute temperature of organs on a 3D plant model are general-purpose developments. Production rules are more specific to the description of the growth and geometric structure of maize. When genotype-dependent, physiological parameters are estimated for the DEA maize cultivar.

L-SYSTEM IMPLEMENTATION

Principle of L-system modelling

General background in L-system modelling of plants has been described in several papers, and one can refer to Prusinkiewicz *et al*. (1997), and references therein, for a general overview. Only the minimum information required to understand the way we used L-systems will be given here.

The plant is viewed as a string of modules (a module being any structural unit repeated in the global structure, here apices, leaves and internodes). The string encodes the plant as an ordered succession of words, representing the modules, and brackets, indicating the beginning and end of ramifications. This bracketed-string notation allows coding of any structure with a 1D topology, i.e. a ramified structure. Development of the plant is encoded as a parallel rewriting process, that transforms the modules into new modules at each time step, starting from an initial state called the axiom. The rewriting process analyses the string of modules and replaces each module where a production rule applies by the appropriate result. Rewriting allows the modelling of recursive processes such as creation of axillary meristems by a father meristem. Parallelism is also important because it makes it possible to deal with processes occuring simultaneously in different parts of the structure.

The transformations are defined by the production rules and can be quantitative or qualitative, to feature growth and development. For instance, if *Am* is a module representing the apical meristem (the apex) the production rule

$$Am \rightarrow I[Axm][L]\,Am \qquad (1)$$

describes the production by the apex of a growth unit consisting of an internode *I*, an axillary meristem *Axm* and leaf *L*. The symbols [and] denote the beginning and the end of a ramification, respectively.

Quantitative evolution of the modules is described through evolution of parameters. For instance, the elongation of a leaf during a time step may be described through the production rule:

$$L(l) \rightarrow L(l + \mathrm{d}l), \quad (2)$$

where l denotes the length of the leaf and $\mathrm{d}l$ the length increment.

Modules may have parameters corresponding to variables involved in physiological processes and others parameters to describe their geometric aspect. In previous example the parameter l may be used for both purposes. Any geometrical parameter such as dimension, or angle can be associated with the modules and manipulated by the production rules. Geometric representation is based on a logo-style turtle interpreter that recognizes a set of reserved modules present within the string as shapes to draw.

Calculations using the 3D structure, such as the physical exchanges of mass and energy with the environment, cannot be performed using production rules. Instead they have to be performed by external programmes, and the specific point here was to develop a bi-directional communication interface. Our work was done independently of Mech and Prusinkiewicz (1996) who developed an interface with environment using their CPFG software for similar purposes. In their case, invoking external programmes and exchanging information was done within the frame of L-system, by a special query module, able to export information to an external programme, run it, and collect information from it. Our implementation will be discussed below.

Graphtal and the bi-directional communication with the environment

The L-system Graphtal was developed by Streit (1992), who mentions that it mimics the main traits of an earlier version of CPFG described in James, Hanan and Prusinkiewicz (1993). The features of Graphtal that were most important for our implementation are: (*a*) the use of brackets as in eqn (1) to define ramifications, and use of parameters as in eqn (2); (*b*) the use of condition and probability, for instance:

$$L(l): l < l_{\max} \rightarrow L(l + \mathrm{d}l)$$

makes a leaf elongate only if the length is less than a predefined length $l_{\max}$.

$$Am(n) \rightarrow (1 - n/10)\, I[Axm]\,[L]\, Am(n+1)$$
$$\rightarrow (n/10)\, AmR$$

makes the apex either produce a growth unit with a probability $P = 1 - n/10$, or change to reproductive stage AmR, with a probability $P = n/10$; here n is incremented at each time step and thus represents the number of vegetative growth units that have already been created; (*c*) the use of basic logical (and, or, etc) and algebraic $(+, -, /)$ operators and a set of pre-defined mathematical functions (sin, cos, log, exp, etc); (*d*) the use of tables to define sub L-system and the possibility of defining macro-instructions. Both improve the clarity of the algorithmic implementation; (*e*) use of global variables, recognized in all tables or local variables specific to one table; (*f*) the use of a set of geometrical modules leading the logo-style turtle: move, rotate, draw a line, a cylinder, a sphere or a polygon. The geometrical data generated by the standard version of Graphtal contains no topological or process related information, it consists only of co-ordinates of points, polygons, etc, associated with a colour.

From Graphtal to the environment. A feature very useful to the bi-directional communication was to structure the model using two sets of rules, the first corresponding to physiological processes and the second to geometric drawing. Production rules applied only for geometric interpretation are called homomorphisms. This distinction improved clarity and versatility in the model development, and facilitated the implementation of communication with the environment described below.

Homomorphisms concentrate on 3D representation and can be applied at time intervals different from those used in the application of the physiological rules. We modified Graphtal so that the output of the homomorphisms were written in a specific file, denoted further as the 'geometric output file'. A new, pre-defined module $W(p_1, p_2, \ldots p_n)$ was incorporated into Graphtal, that writes the set of real or integer parameters $(p_1 \ldots p_n)$ in the geometric output file. W can be invoked each time a homomorphism is applied, to write a set of parameters that is to be attached to that geometric object. The first parameter p_1 is a label that allows the external procedures to recognize the module to which the polygon belongs and its topological location in the plant. Other parameters can be any value calculated through the production rules (e.g. age of module, chlorophyll concentration, etc) and useful to the external models. Finally, this procedure was also extended by creating an additional file, denoted further as the 'information file', containing only the subset of modules expecting information from the environment.

Isolating the geometric aspects allows the string representing the physiological description to remain quite simple. At each time step the L-system model simply integrates the local physiological processes, and converts the result into files for external models.

From the environment to Graphtal. External modules perform calculations from the geometrical output file, using the process-related information, such as exchange properties written in this file by the module W. The results of these external calculations are written in a file, using the same identification label to define the module, then the result of the associated calculation. The set of output modules of the external procedures is not necessarily the same as the set of input modules. For instance, the apex temperature model uses the geometrical information relative to a whole canopy, and provides only the list of meristems with their temperature.

An external string rewriting procedure was developed that enables the string coding for the 'physiological description' to collect the results of the external modules. It uses three input files: the first is the string of modules describing the plant, the second is the output file from the external procedure, and the third is the information file that

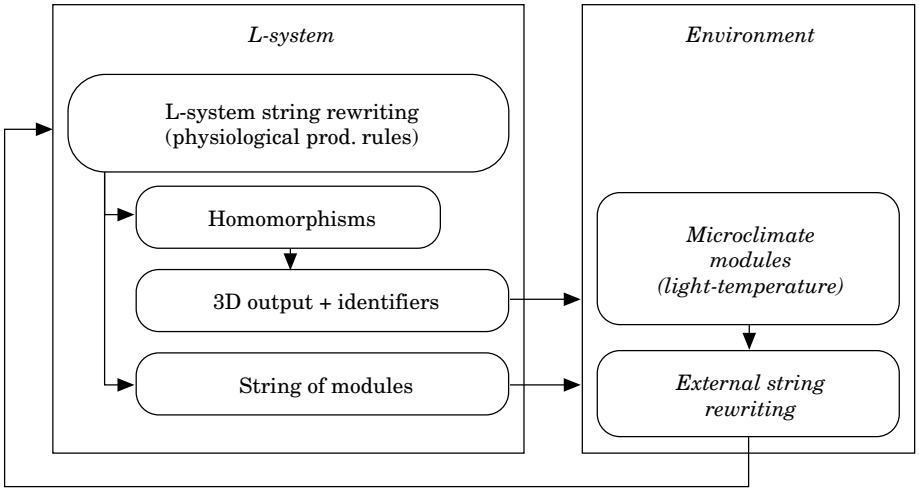


FIG. 1. Diagram showing the four stages of calculation comprising one time step, using bidirectional communication between Graphtal and the environment.

defines the modules and parameters to be rewritten. Then for each occurrence of such a module in the string, the current value of the corresponding parameters is replaced by the value of these parameters calculated by the external procedure.

Running one time step. A complete iteration of the model using the bi-directional communication facility consists of four stages summarized in Fig. 1: (*a*) production rules describing the physiology are applied to a current string of modules; (*b*) homomorphisms are applied to compute the new 3D structure; (*c*) external subroutines use the 3D structure to compute new values of environmental variables; and (*d*) external string rewriting incorporates these values in the L-system string.

PHYSIOLOGICAL RESPONSE OF GROWTH OF MAIZE TO TEMPERATURE

Modular description of maize

Our model focuses on the description of growth of the aerial vegetative structure of maize, during the period from the initiation of leaf six to the full development of the vegetative structure. The first four or five leaves are already present in the embryo (Messiaen, 1963; Juget, Derieux and Duburcq, 1986). The development of the embryo from sowing to seedling emergence involves different processes, which are not considered in this work. Initiation of leaf six corresponds to seedling emergence (Warrington and Kanemasu, 1983; Kiniry and Bonhomme, 1991). Lengths of leaves one to five at seedling emergence are supposed to be genotype-constants, used as inputs to the model. We focused on the development of the main stem and leaves, neglecting tillering which is a phenomenon of minor importance in modern varieties of maize grown under standard field conditions.

The overall structure of the model is based on the general knowledge of leaf and stem expansion for grasses (see, for example, Grant, 1989; Kiniry and Bonhomme, 1991). The starting point in architectural modelling is the definition of modules faithful to the botanical structure of the plant and allowing a description of ontogeny. We defined three main modules with respect to the age and the topological position of the meristem from which they originate: (1) the apex module, denoting the apical meristematic region of the stalk, generating other lateral meristematic regions; (2) the leaf module, that originates from a lateral primordium. We do not distinguish the sheath and blade as distinct modules because the continuous growth of the whole leaf results from the activity of a single meristematic region; and (3) the internode module, that originates from a meristematic region of the same age as the leaf, but later separates from the leaf meristematic region because of intercalary growth (Morrison, Kessler and Buxton, 1994).

The apical meristem

The production rules associated with the apex describe successive initiation of phytomers, consisting of two other modules: an internode and a leaf. No axillary bud is considered, given the framework defined above. The production of phytomers stops when the apical meristem enters its reproductive stage and initiates the panicle.

Initiation of leaves and internodes. Under constant daily temperatures, the rate of production of leaf primordia by the apex was constant (Warrington and Kanemasu, 1983*a*; Lejeune and Bernier, 1996) or slightly higher (Thiagarajah and Hunt, 1982) or lower (Zur, Reid and Hesketh, 1989) for the last three or four primordia. These differences can be accounted for, at least partly, by the difficulty in measurements or by differences in the way primordial stages were recorded by the authors. The apex temperature is by far the main driving factor, although a sensitivity exists to the photoperiod (Warrington and Kanemasu, 1983*a*), and presumably to some other environmental factors. We introduced into our model a rate of initiation independent of the stage of development and varying with temperature according to the experimental function established by Warrington and Kanemasu (1983*a*) in conditions where other factors were not limiting. It reads:

$$R_p = -0{\cdot}00065 - 0{\cdot}0138T + 0{\cdot}00372T^2 - 0{\cdot}000072T^3 \quad (3)$$

This relation is consistent with the experimental data of Thiagarajah and Hunt (1982), but predicts lower rates than those measured by Zur *et al.* (1989), although the slope in the range 13–26 °C is nearly the same: 0·048 pdd^{-1} (primordia per degree-day, base 8 °C). Kiniry and Bonhomme (1991) reported that this slope depends on the genotype, with values ranging from 0·034 to 0·065 pdd^{-1}.

Transition to the reproductive stage. The transition of apex activity from the vegetative to reproductive stage limits the total number of leaves produced. The mechanism for this transition remains unclear. In a controlled environment, a smaller number of leaves was found for temperatures around 18–20 °C (Warrington and Kanemasu, 1983*b*) and for a photoperiod under a threshold value, depending on the genotype (Kiniry, Ritchie and Musser, 1983*a*; Warrington and Kanemasu, 1983*b*). The effects due to changes in photoperiod or temperature are additive (Tollenaar and Hunter, 1983). The increase with temperature outside this range varies among hybrids from 0·13 to 0·44 leaves $°C^{-1}$. The increase with photoperiod also varies among hybrids, from 0 to more than 1·5 leaves per additional hour of daylight (Kiniry *et al.*, 1983*a*; Warrington and Kanemasu, 1983*b*). Sensitivity to temperature and photoperiod occurs only after a given stage of development, which has been expressed in term of days before tassel initiation (Kiniry *et al.*, 1983*b*), sum of temperature (Kiniry *et al.*, 1983*a*) or leaf stage (Tollenaar and Hunter, 1983).

Jones and Kiniry (1986) proposed a model in which the stage of sensitivity to photoperiod and temperature is seen as a phase of induction of the reproductive apex, that starts after the plant has encountered a given (genotype-dependent) number of degree-days. During the induction phase, the apex continues to emit new primordia every 21 degree-days. The duration of the phase is independent of temperature and increases linearly with the photoperiod, with a slope depending on the genotype. Finally the total leaf number for a given genotype varies according to the number of primordia initiated during the inductive phase. However this model does not reproduce the increase in number of leaves for temperatures lower than 18 °C.

Here we included the parameterization proposed by Grant (1989): the inductive stage starts after initiation of a given number of primordia. This number represents the minimum number of leaves for that genotype. The number of additional primordia is the sum of a function of photoperiod, *PPD*, and a function of temperature, *TMP*, which are evaluated daily as an integral from the beginning of the inductive phase:

$$PPD = \frac{1}{t}\int_0^t \max \quad [0, RI(DL - 12{\cdot}5)]\,\mathrm{d}x$$

$$TMP = \frac{1}{t}\int_0^t (13{\cdot}6 - 1{\cdot}89T + 0{\cdot}081T^2 - 0{\cdot}001T^3)\,\mathrm{d}x. \quad (4)$$

Here DL is the day length (h), RI a parameter of sensitivity to photoperiod (expressed in leaves h^{-1}, can be 0), and T (°C) is the apex temperature. The inductive stage ends when the number of additional leaves reaches the current value of $PPD + TMP$.

Elongation of leaves and sheath

Processes at the local level. The production of the leaf lamina and sheath occurs in a single region of cell division, at the base of the leaf (Skinner and Nelson, 1995). The cells progress to a region where extension predominates and finally reach a maturation region, about 9 cm from the base of the leaf (Meiri, Silk and Läuchli, 1992; Ben Haj Salah and Tardieu, 1995).

In the region of cell division, the area producing the lamina and sheath is distinct from the very beginning of leaf elongation and separated by a non-meristematic line that corresponds to the future position of the ligule. The position of this line varies with the time, the area of lamina production being predominant at the beginning of leaf growth, and the area of sheath production being predominant at the end. Data presented by Sharman (1942), Hesketh *et al.* (1988), Khouja (1990), Robertson (1994) and Ben Haj Salah (1996) indicate that the sheath reaches macroscopic size when the lamina is about 80 % of its final length, and the sheath is about 30 % of its final length (i.e. 4 to 7 cm) when elongation of the lamina ceases.

The growth rates of sheath and lamina are controlled primarily by temperature (and can be modified by environmental factors such as water potential and salinity). The transition from lamina to sheath growth does not coincide with a modification in the rate of extension of the leaf (Khouja, 1990), so for the same environmental conditions the growth rate seems identical for both sheath and lamina.

According to Ben Haj Salah and Tardieu (1995), the rates of cell production and cell extension depend in the same way on the temperature, so that variations in temperature change the leaf growth rate without modifying the size of cells in the mature leaf. Thus we did not try to mimic this process of cell division and expansion in detail, but used 'leaf' modules to describe the elongation resulting from this process, with the same temperature response for sheath and lamina. As an approximation to results cited above, we considered that the growth of the leaf was first entirely that of the lamina and then that of the sheath. The parameterization for the kinetics of leaf elongation is described below.

Parameterization of leaf elongation. The growth curve of a leaf represents three phases: (1) a period of quasi-exponential growth, corresponding to the development of the growth zone from the primordia; (2) a period of linear growth, corresponding to a nearly stationary activity of the growth zone; and (3) a period of decreasing growth rate. We refer to Arkebauer and Norman (1995*a*, *b*) and Arkebauer, Norman and Sullivan (1995) for a detailed analytic model.

Much data are available for the linear growth rate calculated from measurements of the distance between the leaf tip and the whorl. However this distance results from both the growth of the lower internodes and the growth of the leaf (Gallagher, 1979). Analysis of kinetic curves obtained from direct measurement of leaf elongation of maize by Grobbelaar (1963), Hesketh and Warrington (1989), Khouja (1990) and Meiri *et al.* (1992) shows that growth rate in phase two is independent of leaf rank, and the duration of phase two depends on leaf rank. This

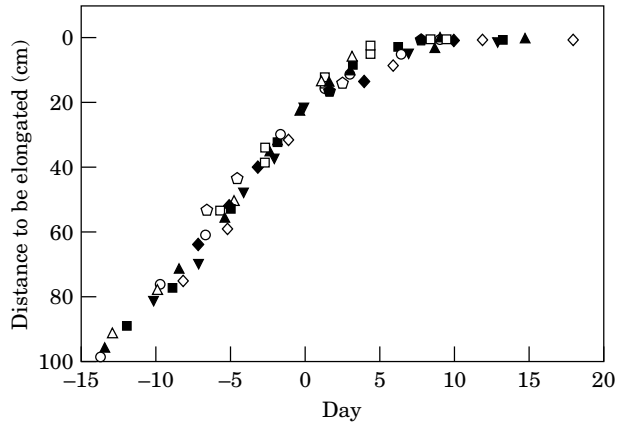


FIG. 2. Evaluation of the model of elongation of a maize leaf by a broken line segment. Data reprinted from Hesketh and Warrington (1989). Symbols are for leaves of rank four to 15. Original data were translated along the x-axis so that the time origin for any leaf is the moment where the leaf is 20 cm shorter than the final length.

explains most of the variation in final leaf length within a plant. The idea of a growth rate independent of leaf rank is also supported by measurements of linear growth rate of DEA corn leaves five to ten by Khouja (1990) and Ben Haj Salah (1996). It is consistent with observations by Arkebauer and Norman (1989) and Ben Haj Salah (1996) that the size of the elongation region is independent of the rank of the leaf.

Unfortunately, nearly no data are available on phase one, during which growing organs are not apparent. Concerning phase three, it can be concluded from the kinetic curves of leaf elongation mentioned above, that only the elongation of the last 5 to 10 cm of tissue is done at a rate significantly lower than that of the linear phase. Finally, we approximated the growth curve of a leaf by a linear segment, depending on three parameters: (1) the delay between initiation and beginning of elongation; (2) the growth rate; and (3) the duration of linear growth. The second parameter is supposed to be independent of leaf rank.

We evaluated this parameterization using measurements by Hesketh and Warrington (1989) of the length of the visible fraction of the leaves, as a function of time. To compare the growth curves of individual leaves, we translated original data along time and distance axes: in Fig. 2, the y-axis represents the distance to be elongated before the end of leaf growth, and the time origin (on x-axis) was calculated for each leaf as the moment when it was 20 cm less than its final length. We did not report measurements taken before the beginning of phase two. As mentioned above, the beginning of elongation of the internode may interfere with the measurement of the end of the leaf growth curve. This could result in slight overestimation of elongation during phase three, at least for the top-most leaves. Even in these conditions, the end of elongation appears to occur quite abruptly. Finally, Fig. 2 shows that the growth of all leaves during phase two and three is well approximated by considering a growth rate independent of leaf number and constant to the end of elongation.

Leaf growth rate as a function of temperature. A large amount of work has been devoted to studying the influence of temperature on leaf elongation. However methods are quite heterogeneous, involving either real growth rate (elongation of the leaf) or apparent growth rate (calculated from distance of leaf tip to the whorl), measured either in the linear phase or averaged over the totality of leaf growth. The temperature is generally that of the air, or the soil; rarely that of the apex. Despite this heterogeneity, there is a general agreement that growth rate increases quasi-linearly with increasing temperature in a range $[T_{base}, T_{opt}]$, then decreases down to zero for temperatures higher than T_{max} (Ong and Baker, 1985). Since very few experiments have been performed for temperatures significantly higher than T_{opt}, the behaviour in this region is most uncertain.

Estimates of T_{base} when air temperature is measured are generally between 6 and 10 °C, the value $T_{base} = 8$ °C being most widely used. Values reported for T_{opt} are generally between 30 and 32 °C. Values proposed for T_{max} are around 50 °C. We found only two studies in which temperature was measured directly on the plant, both concern the cultivar DEA. Soontornchainaksaeng (1995) measured temperature between sheaths close to the apex and Ben Haj Salah (1996) measured the temperature of the apex directly. For the region of increasing growth rate with temperature, we used the parameter proposed by Ben Haj Salah (1996):

$$v_l = 0{\cdot}564\,(Tm - 9{\cdot}8) \tag{5}$$

where v_l is the growth rate in cm d^{-1} and Tm is meristematic temperature. The value $Tm_{base} = 9{\cdot}8$ °C is higher than the usual value of 8 °C, which may result from the use of apex temperature measurement. The slope $dv/dTm = 0{\cdot}564$ is close to the value reported by Soontornchainaksaeng (1995) and other authors. We used $Tm_{opt} = 31$ °C, and $Tm_{max} = 50$ °C, consistent with results obtained by Soontornchainaksaeng (1995), and close to estimates by authors using air temperature.

Duration of growth and final leaf size for leaf laminae. Several mechanisms have been proposed to explain the end of activity of the elongation zone, such as competition between growing organs (Schoch, 1974; Keating and Wafula, 1992), response to a light stimulus (Begg and Wright, 1962), or co-ordination by the apex (Tesarová, Seidlová and Nátr, 1992; Skinner and Nelson, 1994) but there is no clear experimental evidence for any of these mechanisms.

Analysing the results of Grobbelaar (1963), Hesketh and Warrington (1989), Plénet (1995), Soontornchainaksaeng (1995) and Ben Haj Salah (1996), the final length of the leaf is stable for a wide range of temperatures, typically between 16 and 28 °C. All these authors observed a reduction in final length for temperatures above 30 °C, which also corresponds to a decrease in growth rate. Few results are available for temperatures lower than 16 °C: Khouja (1990) observed a reduction in leaf length for an air temperature of 14 °C, whereas Grobbelaar (1963) found no effect on leaf size for a soil temperature of 15 °C. Here we assumed that the end of the elongation of a leaf occurs after a given physiological growth duration, expressed in degree-days. This implicitly assumes that the concept of degree days represents the physiological aging of leaves, even at high temperature when the growth rate decreases due to lower cell extension.

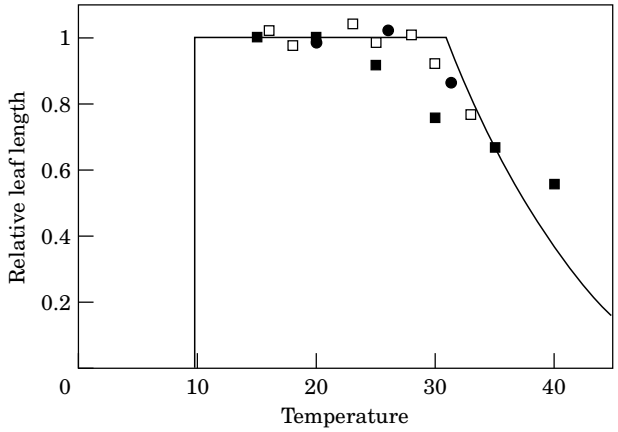


FIG. 3. Influence of temperature on final length of laminae. The relative leaf length is the ratio of the length to that obtained for plants grown at $T < T_{opt}$. The line represents the model; symbols are for data from (□) Hesketh and Warrington (1989), $T = T_{air}$; (■) Grobbelaar (1963), $T = T_{soil}$; and (●) Soontornchainaksaeng (1995), $T = T_{apex}$.

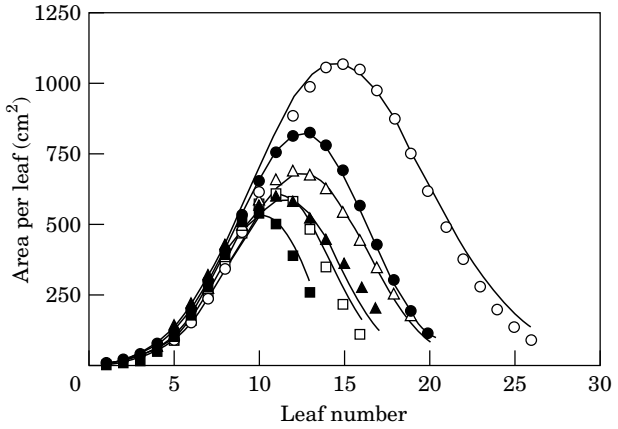


FIG. 4. Area of lamina as a function of leaf rank, for genotypes of maize with different numbers of leaves. Lines represent area calculated from eqns (6) and (7), Y_M and Nt are as input. Symbols are for data by (■) Keating and Wafula (1992); (▲) Muchow and Carberry (1990); (△) Plénet (1996); (□ and ●) Dwyer *et al.* (1992); (○) Cooper (1979).

This calculation results in a final leaf length independent of temperature below 31 °C and a reduced size at higher temperatures. Figure 3 shows that final lengths calculated for various temperatures compare quite well with published measurements.

Calculating leaf growth duration as a function of leaf rank requires an estimation of the profile of length of mature leaves for temperature in the range 16–28 °C. A large amount of data is available in the literature concerning final area of maize laminae, but few authors present data relative to final length. There is some systematic variation in the ratio between lamina width and length according to leaf rank (Greyson, Walden and Smith, 1982), however this is a second order term compared to variations in leaf length, so we approximated profiles of lamina length as a root-square function of profiles of lamina area.

The area of the lamina increases from leaf one to the largest leaf, which is in most cases, within one rank of the ear leaf, and then decreases. The bell-shaped curve, (Dwyer and Stewart, 1986; Dwyer *et al.*, 1992) has proved to be an appropriate parameterization of the vertical profiles of lamina area. It can be written:

$$Y(n)/Y_M = \exp[a(n/n_M - 1)^2 + b(n/n_M - 1)^3] \quad (6)$$

where $Y_{(n)}$ is the area of lamina of rank n, Y_M and n_M are the area and the rank of the largest lamina, respectively, and a and b are two parameters. The parameters a, b, and n_M can be reasonably estimated from the total number of leaves, Nt (see also Muchow and Carberry, 1989; Keating and Wafula, 1992). We compiled data for 22 genotypes published by Cooper (1979), Thiagarajah and Hunt (1982), Dwyer and Stewart (1986), Muchow and Carberry (1989), Keating and Wafula (1992), Dwyer *et al.* (1992) and Plénet (1995) and found:

$$\begin{aligned} n_M &= +5{\cdot}93 + 0{\cdot}33Nt \quad (r^2 = 0{\cdot}74) \\ a &= -10{\cdot}61 + 0{\cdot}25Nt \quad (r^2 = 0{\cdot}54) \\ b &= -5{\cdot}99 + 0{\cdot}27Nt \quad (r^2 = 0{\cdot}55) \end{aligned} \quad (7)$$

Figure 4 shows that these approximations reproduce quite faithfully the profiles of leaf area measured for a wide range of genotypes with a total number of leaves between 12 and 30. For the whole data set (22 genotypes, 408 leaves), the correlation coefficient r^2 between calculated and measured leaf area was 0·96.

The only unknown parameter is Y_M, the area of the largest leaf. Data found in the literature show significant variability, but no systematic variation with the number of leaves when different genotypes are considered together. For a given genotype and a given final number of leaves, maximum leaf area seems to be quite stable, in the absence of hydric and thermal stress. On the other hand, we found few data relating the maximum size of leaves to the number of leaves for a given genotype (e.g. when the number of leaves produced in the inductive period is varied). As a rough approximation, our calculation takes into account a linear increase of maximum leaf area with total leaf number.

Finally, physiological duration of growth $D^*(n)$, of a leaf of rank n is calculated as:

$$\begin{aligned} D^*(n) &= L(n)/v^* \\ L(n) &= L_M \exp\left[\frac{a}{2}(n/n_M - 1)^2 + \frac{b}{2}(n/n_M - 1)^3\right] \\ L_M^2 &= L_{M\,min}^2 + K(Nt - Nt_{min}) \end{aligned} \quad (8)$$

Here, v^* is a constant, equal to the value of the growth rate, expressed in physiological time for temperatures lower than Tm_{opt}; according to eqn (5), $v^* = 0{\cdot}564$ cm dd^{-1} for our DEA cultivar. $L(x)$ is the final leaf length. L_M (cm), the length of the largest leaf varies with the number (Nt) of leaves produced by the apex, so that Y_m increases linearly with Nt. Nt_{min} is the minimum number of leaves for the genotype, i.e. the number of primordia induced at the start of the inductive period ($Nt_{min} = 14$ for DEA). We estimated $K = 24$ cm² per leaf, based on data produced by Allison and Daynard (1979) and by Muchow and Carberry (1989). $L_{M\,min}$ is a length characteristic of the genotype; it was

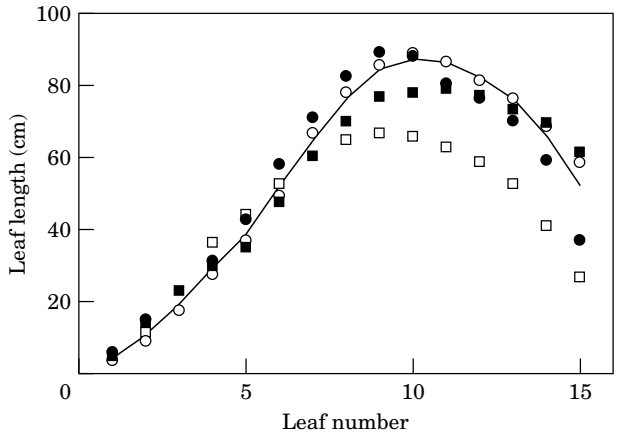


FIG. 5. Length of lamina of maize DEA cultivar, as a function of leaf rank. Line represents lengths calculated for $Nt = 15$ and $T < T_{opt}$. Symbols are for experimental data by Ben Haj Salah (1996), for different dates of sowing.

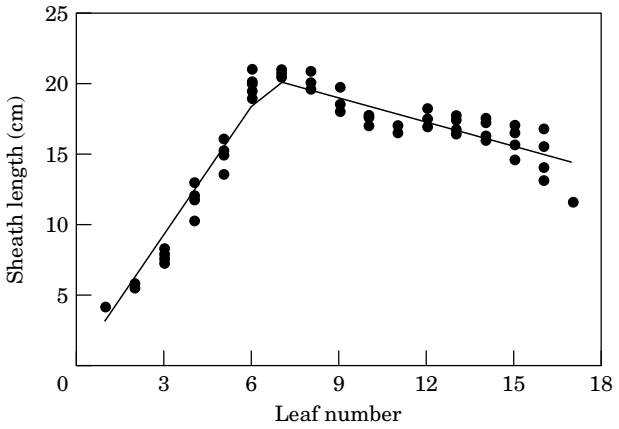


FIG. 6. Sheath length of maize DEA cultivar, as a function of leaf rank. ●, Measurement on five plants. The line is the fitted curve described by eqn (10).

adjusted to 90 cm for the DEA cultivar in irrigated field conditions following data from Ben Haj Salah (1996) and Prévot *et al.* (1991). The mean value of L_M is estimated with good accuracy, whereas the coefficient K is only representative of a trend. However, in the case of the DEA cultivar, the final number of leaves varies within a quite narrow range, typically from 15 to 17, so that an accurate estimate of K is of limited importance.

The physiological age of a leaf is calculated at any time t as:

$$a^* = \int_{t_i}^{t} \mathrm{d}t^* = \int_{t_i}^{t} (Tm - Tm_{\mathrm{base}})\,\mathrm{d}x \qquad (9)$$

where t_i is the date on which leaf elongation begins and $\mathrm{d}t^*/\mathrm{d}t = Tm - Tm_{\mathrm{base}}$ represents the rate of the physiological time course of the leaf meristem. Leaf elongation ends when the physiological age, a^*, matches the physiological growth duration D^*. For leaves growing before tassel initiation, Nt is not known and D^* is calculated using $Nt = Nt_{\mathrm{min}}$. This choice is somewhat arbitrary but, for an early flowering genotype such as DEA, it concerns only the embryonic leaves. Moreover, it is qualitatively consistent with figures produced by Allison and Daynard (1979) and by Muchow and Carberry (1989) that show no noticeable variations in the size of the lower leaves for plants of a given genotype growing under environmental conditions leading to different final number of leaves.

Figure 5 compares simulated profiles of leaf length to data produced by Ben Haj Salah (1996), for the DEA cultivar. Apex temperature was not measured, thus we performed the calculation for a temperature below the threshold leading to a reduction in size. The measured leaf lengths are quite stable for four experiments and are reasonably matched by the simulation. One experiment resulted in significantly shorter leaves, which, according to Ben Haj Salah (1996), may be due to hydric stress.

Duration of growth and final size of leaf sheaths. The end of the growth period of the leaf sheaf is based on the criterion of a defined growth duration, expressed in physiological time, similar to the leaf lamina. Since sheath growth rate varies with temperature in the same way as lamina growth rate, this results in the same response to temperature for final sheath size as for final lamina size. This is consistent with results by Khouja (1990), showing a constant ratio between the length of the leaf and the length of associated sheath when temperature was varied between 14 and 30 °C.

Whereas the response to temperature seems the same for sheath and lamina, profiles of sheath length along the plant are very different to profiles of lamina length. We measured the length of sheaths on a DEA corn for a classic agronomic treatment (sown on 2 May 1996 at Grignon, France, with a population of 100 000 plants ha^{-1}). Results are presented in Fig. 6. They show a regular increase in final length of successive sheaths for the first six or seven leaves, then a moderate decrease for higher leaves. There is a depression around leaf ten or 11, corresponding to the ear leaf. This shortening may be associated with the strong sink for nutrients caused by the ear (Morrison *et al.*, 1994), but remains moderate for a crop growing under normal agronomic conditions. These patterns are consistent with data previously published: Robertson (1994) observed maximum sheath length at leaf six or seven for different maize genotypes with a final number of leaves between 16 and 20. Grant and Hesketh (1992) observed the same feature for one genotype with $N_t = 20$, grown under varied populations.

Finally, the physiological duration of sheath elongation, $G^*(n)$, was adjusted through two segments of a line, with coefficients calculated so that final lengths adjust our measurements for temperatures around 20 °C.

$$\begin{aligned} G^*(n) &= (1/v^*)\,3077n && \text{for } 1 \leqslant n \leqslant 6 \\ G^*(n) &= (1/v^*)\,(2408 - 0{\cdot}569n) && \text{for } n \geqslant 7 \end{aligned} \qquad (10)$$

Delay between initiation of primordia and beginning of leaf elongation. Because observation is very difficult, few direct observational data are available on the development of primordia. On the other hand, the synchronism between initiation of primordia and leaf ligule appearance is well

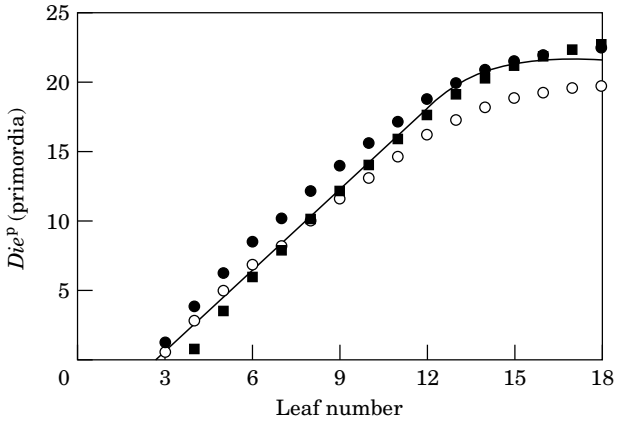


FIG. 7. Delay between initiation and beginning of leaf elongation, as a function of leaf rank. Delay is in plastochron units. The line represents the model (fitted curve). Symbols are for the estimate from experimental data by Cao *et al.* (1988) and Zur *et al.* (1989) at three day/night temperature regimes: 19/14 °C (○); 26/20 °C (■); and 30/24 °C (●).

established. Leaf ligule appearance occurs nearly simultaneously with the end of leaf elongation (Hesketh and Warrington, 1989). We estimated the delay *Die* between primordium initiation and beginning of leaf elongation as the difference between *Dil*, the duration from the initiation of the primordium to the ligule appearance of the leaf and *Del*, the duration of phase two of leaf growth (sheath and lamina).

We used data produced by Zur *et al.* (1989) and Cao *et al.* (1988). They correspond to a single experiment, in which the following variables were monitored for three temperature regimes: (*a*) the dates of leaf initiation and leaf ligule appearance from which we calculated *Dil* as a function of leaf rank; and (*b*) the final area of each leaf, from which we evaluated *Del* as a function of leaf rank, using eqn (4) to calculate the growth rate.

We found that *Die* depended on leaf rank, n, and on the temperature regime. We searched for a physiological time unit that would take into account the temperature effect. When expressed using the physiological leaf age defined by eqn (8), $Die(n)$ still depended on temperature. However, when durations were expressed in plastochronic units, calculated according to eqn (3), the curves for the three temperatures nearly matched (Fig. 7). Die^p representing *Die* expressed in plastochronic units, is given independently of the temperature regime by a linear function of n for $n > n_M$:

$$Die^p(n) = -5{\cdot}16 + 1{\cdot}94\,n \quad (r^2 = 0{\cdot}95) \tag{11}$$

For upper leaves, we adjusted a third order polynomial function with the following constraints: (1) continuity with the linear function at n_M; (2) identical slopes at n_M; (3) slope equal to zero at Nt; and (4) $Die^p\ (Nt) - Die^p(n_M) = 3{\cdot}65$ primordia.

Since our model runs with real time, Die^p needs to be converted in *Die* (real time units). The relation is:

$$Die^p(n) = \int_{t_i}^{t_i + Die(n)} Rp(T)\,\mathrm{d}u \tag{12}$$

where t_i is the date of leaf initiation and $Rp(T)$ is the rate of primordium initiation calculated using eqn (3). Equation (12) is solved numerically as follows: at each time step, the integral $\int_{t_i}^{t} Rp(T)\,\mathrm{d}u$ is evaluated, and the difference, Δ, with $Die^p(n)$ is calculated. If the difference is greater than 0, then the leaf is considered to be in a linear stage of elongation. The difference is converted to real time and the current leaf length is calculated according to the rules for leaf elongation.

Elongation of internodes

The first four or five internodes, supporting the roots, remain very short. Significant elongation occurs only for higher internodes and starts after the apex has formed a tassel (Messiaen, 1963). Plant height, and thus internode length is known to be significantly affected by population density through trophic and photomorphogenetic processes. This is illustrated for example in Grant and Hesketh (1992). However, taking account of these processes is out of the scope of this study, and our parameterization is based on data corresponding to the usual agronomic density (eight plants m^{-2}).

There is no theoretical evidence of a necessary synchronization between the elongation of a leaf and that of the associated internode. However Sharman (1942), Hesketh *et al.* (1988), Grant and Hesketh (1992) and Robertson (1994) related the beginning of internode elongation to the development of the associated leaf. According to their data, we estimated that an internode starts elongating approximately when the sheath has reached 60 % of its final length. This means that typically 5 to 10 cm of the sheath remains to elongate, and thus is strikingly close to the end of phase two in leaf elongation. Given our simple parameterization of leaf growth kinetics, we considered that internode elongation immediately follows leaf elongation.

Growth rate of internodes. We found few data in the literature concerning the rate of elongation of internodes v_i. Robertson (1994) measured a mean rate of elongation of 3 cm per phyllochron in the field independent of the rank of the internode. Morrison *et al.* (1994) monitored the growth rate of internodes and estimated an average growth rate of 9·5 mm d^{-1} for a constant temperature regime of 28 °C during the day and 18 °C during the night. These measurements correspond to ratios of v_i/v_l of 0·14 and 0·13, respectively. However, this is for the wide period of internode growth, v_e whereas most of the elongation takes place in a shorter period at a higher rate. Finally we estimated $v_i = 0{\cdot}18\ v_l$ was representative of the average growth rate during the period in which most elongation occurs. This parameterization implicitly assumes that v_i varies with temperature in the same way as v_l.

Final size and growth duration of internodes. We found few data in the literature regarding the effect of temperature on the final size of internodes. The final height of a plant is not reported to vary significantly within the usual range of temperature (Hesketh and Warrington, 1989). Thus we used the same approach as for leaves, and used a criterion of achieving a defined growth duration expressed in physiological time, to stop the elongation of an internode.

Similarly to leaves, internode growth duration was estimated from measurements of profiles of final length.

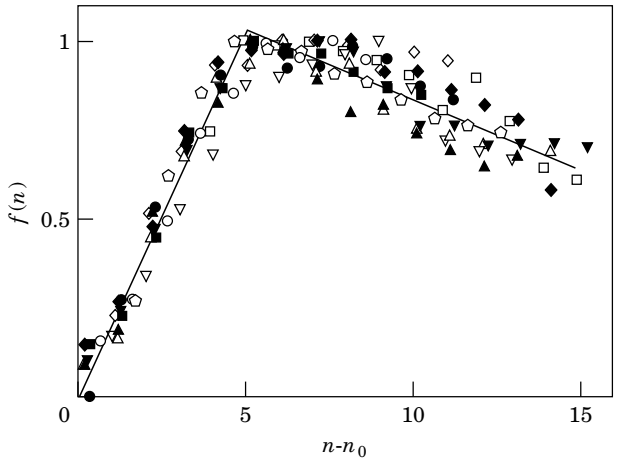


FIG. 8. Normalized internode lengths as a function of the rank of the phytomer. Continuous line is for eqn (14); symbols represent experimental data on 11 genotypes. For each genotype, internode lengths were normalized with respect to the longest internode and n_0 was estimated using linear regression. Data from: (□) Morrison *et al.* (1994); (■, ○) Ephrath *et al.* (1994); (●, △, ▲, ▽, ▼, ◇) Robertson (1994); (◆) Hanway (1970); and (⬠) our own data on the DEA cultivar.

Parameterization of the profile of internode length is based on data published by Hanway (1970), Ephrath, Hesketh and Alm (1994), Morrison *et al.* (1994) and Robertson (1994), as well as our own measurements on the DEA cultivar. The whole data set represents 15 genotypes. It shows, for usual agronomic conditions, a linear increase of internode length for the first five internodes that elongate, then a moderate decrease with rank for higher internodes. Maximum length depends on the genotype (from 12 and 24 cm, with a mean of 20 cm in the data set cited above), but does not correlate with total number of leaves produced. We finally expressed the final length of internodes by:

$$I(n) = I_M f(n) \tag{13}$$

where $I(n)$ is the final length of internode n, I_M is the final length of the longest internode (23 cm for DEA) and $f(n)$ is a broken line function which we adjusted on 11 genotypes from the data set cited above (Fig. 8).

$$\begin{cases} f(n) = 0{\cdot}202\,(n-n_0) & \text{for } n_0 \leqslant n < n_0+5 \quad (r^2 = 0{\cdot}81) \\ f(n) = 1{\cdot}233-0{\cdot}040\,(n-n_0) & \text{for } n \geqslant n_0+5 \quad (r^2 = 0{\cdot}76) \end{cases} \tag{14}$$

where n_0 is the rank of the higher internode that does not elongate. Following Grant and Hesketh (1992), we considered that an internode elongates if the beginning of its potential period of growth (i.e. the ligule appearance of the corresponding leaf) occurs after tassel initiation.

Leaf width and stem diameter

Leaf width and stem diameter are known to be much more sensitive than length to trophic conditions (Grant and Hesketh, 1992; Soontornchainaksaeng, 1995). Since trophic conditions are not yet introduced in our model, we considered a stem diameter B independent of growth conditions, with a linear decrease of diameter with rank, adjusted using our measurements for the DEA cultivar:

$$\begin{cases} B(\text{cm}) = 2{\cdot}3 & \text{for } n \leqslant 7 \\ B(\text{cm}) = 3{\cdot}3-0{\cdot}17n & \text{for } n > 7 \end{cases} \tag{15}$$

Heteroblasty between leaves was not considered, and leaf width was considered proportional to leaf length. Parameterization of leaf shape is detailed below.

Geometric representation of maize

The production rules associated with physiological process predict the length of internodes, sheaths and blades as a function of time and apex temperature. Geometric representation requires us to define the other dimensions and the orientation of plant organs. Internodes were approximated by cylinders, with a diameter decreasing from the bottom to top of the plant according to eqn (15). The representation of leaves is based to a large extent on the work by Prévot *et al.* (1991).

Leaf width. The shape of a leaf developed on a plane surface is described by a relation between leaf width and position on the midrib. There is no secondary growth of tissue in grasses, thus the shape of the lamina at any stage of growth is the shape of the fully developed leaf, truncated at the level of the cell elongating zone. We used the parameterization of the shape of a fully developed leaf proposed by Prévot *et al.* (1991).

$$\frac{w}{W} = -2{\cdot}50\left(\frac{u}{L}\right)^2 + 1{\cdot}84\frac{u}{L} + 0{\cdot}66, \tag{16}$$

where u is the distance to the ligule and L the total length of the lamina; w is the width at point u and W is the maximum width of the lamina. This parameterization corresponds to a shape factor of 0·748 between leaf area Y and leaf dimensions: $Y = 0{\cdot}748WL$. The maximum width of a leaf was approximated from its length according to Prévot *et al.* (1991): $W = 0{\cdot}106\ L$.

Equation (16) was re-expressed as a function of distance to the leaf tip instead of distance to leaf ligule. Thus, the width of the leaf portion emerging from the cell division zone can be calculated at any stage of leaf growth (Fig. 9). The width of the sheath is taken as the width of the leaf at the level of the ligule.

Leaf curvature. Prévot *et al.* (1991) proposed a 3D representation of a fully developed maize leaf based on a parameterization of the curvature of the midrib, supposed to belong to a vertical plane. They described the midrib as an arc of parabola for the ascendant part, followed by an arc of ellipse for the descendant part. Use of two curves allows for a possible discontinuity in the curvature of the midrib. They established the distribution of the seven parameters involved in their parameterization for fully developed leaves of plants grown under standard field conditions.

Here, we used this parameterization, and approximated the shape by ten plane polygons. We described the evolution of the shape of the midrib of a growing leaf using a very simple scheme, considering three stages (Fig. 9): in the first

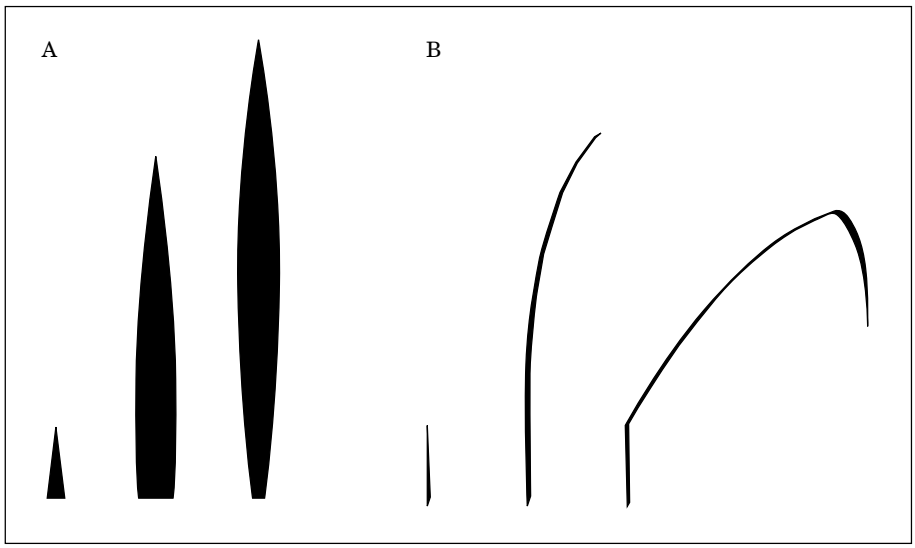


FIG. 9. Shape of a maize leaf, simulated at different stages of growth (A). Curvature of the midrib, simulated at the same stages of growth (B).

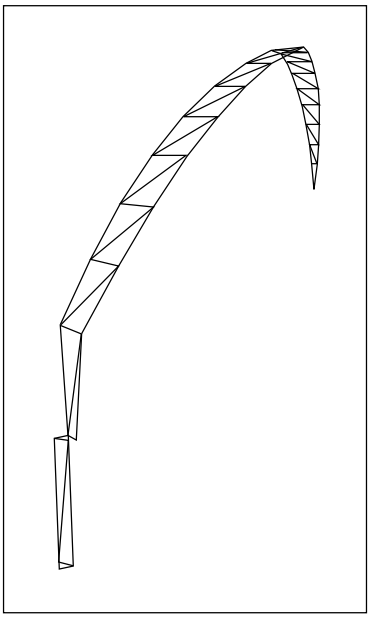


FIG. 10. Representation of a phytomer by a set of triangles. The internode is approximated with eight triangles, the sheath with two triangles, and the blade with 29 triangles.

stage, the midrib is rectilinear and the leaf elongates vertically. This stage corresponds to the growth of the leaf inside the whorl and ends when the tip of the leaf reaches the upper part of the sheath of the former leaf; in a second stage, the midrib acquires the curvature of the fully developed leaf. However, the angle of insertion of the lamina remains vertical, the midrib prolonging the sheath without discontinuity; and at the stage of ligule appearance, the angle between the vertical sheath and the beginning of the lamina is given its final value.

Leaf azimuth. The first five leaves of a maize plants lie in a single plane, whose azimuth is randomly distributed within a field (Girardin, 1992). For upper leaves, azimuth generally differs from that of this initial plane, with a distribution depending on the initial orientation of the plane and on the rank of the leaves considered (Drouet and Moulia, 1996).

Here, we considered a random azimuth for the plane containing the first five leaves, and a random deviation from this plane depending on leaf rank for upper leaves, based on data produced by Drouet and Moulia (1996) for the DEA cultivar. Deviation angle was randomly drawn, with uniform probability density, in the interval $[-20°, +20°]$ for leaves of rank six to eight and in the interval $[-60°, +60°]$ for leaves of higher rank.

Triangulation. To enable visual representation and computation of microclimate described above, the stems and leaves are approximated by a set of triangles (Fig. 10). As mentioned previously, each triangle is given a label enabling us to define, in a unique way, the phytoelement to which it belongs.

Calculation of apex temperature

Calculation of apex temperature is based on the work by Cellier *et al*. (1993) and later developments by Guilioni and Cellier (pers. comm.). Only a brief description of the model is given here, further information can be found in the paper cited above. The model was designed to compute the temperature of the apex in the first stages of growth (typically, for $lai < 0{\cdot}5$). It simulates the temperature of the portion of stem at the level of the apex, using an energy budget equation that takes into account the fluxes of latent heat LE, sensible heat H, and net radiation Rn. The inputs to this local energy budget are: wind speed at the level of the apex and at the level of soil; air temperature at the level of the apex and temperature of the soil; short-wave radiation intercepted by the apex; stomatal resistance of the sheath at the level of apex; and albedo of the soil (a_s) and of phytoelements (a_v).



FIG. 11. Hand-drawn (A) and simulated (B) plants of DEA cultivar at four stages of growth: six, eight, 12 and 13 ligulate leaves on the main stem. Drawings are reprinted from Ledent *et al.* (1990).

The characteristics of the atmosphere at the level of the apex are calculated from data monitored routinely by meteorological stations, using standard parameterization of vertical profiles of wind speed and temperature. The authors found that stomatal resistance could be satisfactorily estimated through a relation with incoming solar radiation on the sheath at the level of the apex. Soil temperature is input into the model.

In Cellier *et al.* (1993), the short-wave radiation, *Rs*, at the level of the apex is calculated as the sum of three components: the direct solar radiation *Rds*, the downward diffuse sky radiation *Rdd*, and the fraction of downward radiation reflected by the soil, $a_s(Rd+Rdd)$. This approximation neglects shading and scattering by the canopy, which was justified by the stage of development considered here, with $lai < 0{\cdot}5$. However, it was later found (Guilioni and Cellier, pers. comm.) that the apex was shaded most of the time by leaves of the same plant, regardless of canopy *lai*. Finally, an empirical approximation $Rs = 0{\cdot}25\ Rg$ where *Rg* is the global solar radiation above the canopy, was found to be the best approximation of incoming solar radiation at the apex. This illustrates a case where the microclimate at the site of interest is different from the average climate at the same level in the canopy. For our model, we used a projection method to calculate *Rds*, taking into account the exact shading by the simulated 3D structure. Since calculation is fast, *Rdd* was calculated by repeating the process for a set of directions sampling the hemisphere.

EVALUATION OF THE MODEL

We compared model simulations to various data sets presented in the literature. When available, temperature conditions were used as apex temperature in the calculation. In the other cases, a constant temperature of 20 °C was used for the model, and the results were compared to experimental data using a degree-day time scale. All simulations were performed using only two parameters: the size of the largest leaf (L_M) and the total number of leaves (Nt). These two parameters were used instead of minimal number of leaves (Nt_{min}) and corresponding leaf size (L_{Mmin}), which were generally not given in the articles. As a consequence, the number of leaves and the size of the largest leaf became constant throughout the simulation.

Visualization of development

We compared drawings of DEA plants at different stages of growth (Ledent, Henkart and Jacobs, 1990) to plants simulated for the same stages. Figure 11 presents these results for four growth stages (six, eight, 12 ligulate leaves and final development), for which information given by Ledent *et al.* enabled us to define the stage of development with good precision. The general agreement indicates that both the geometric representation and the co-ordination between phytomers are quite realistically described by our model. However, before ligule appearance of leaf eight, simulated plants presented a smaller number of leaves in the whorl, compared to the drawings.

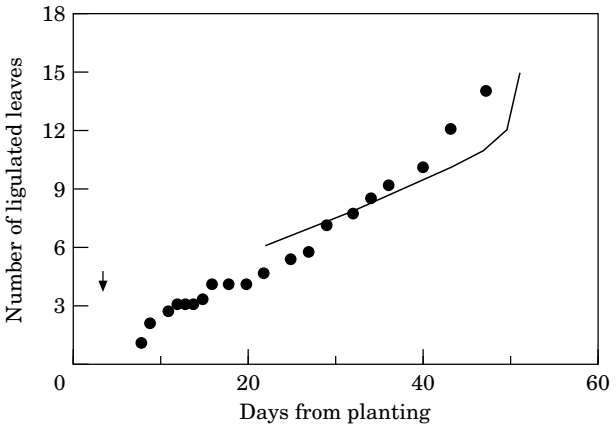


FIG. 12. Measured (symbols) and simulated (line) number of ligulate leaves as a function of time, for the Dekalb XL45 maize cultivar grown at 26/21 °C day/night temperature. The arrow indicates the time of initiation of leaf six. Experimental data from Hesketh and Warrington (1989). Simulations are for $Nt = 15$ and $L_M = 110$ cm.

Development of leaves

Rate of appearance of ligules. Figure 12 presents the time-course of leaf ligule appearance measured on Dekalb XL45 cultivar by Hesketh and Warrington (1989) in a growth chamber experiment, together with simulations by our model. The 26/21 °C day/night temperature regime was taken as apex temperature in the simulations. Hesketh and Warrington (1989) indicated the date of initiation of leaf six, and this was used to calibrate the time origin. The parameters Nt and L_M were set, to 15 and to 110 cm, respectively.

The dates of ligule appearance of embryonic leaves are input parameters in our model and thus are not shown here. Our model reproduces the characteristic increase in ligule appearance rate occurring in later stage of development (Hesketh and Warrington, 1989; Zur *et al.*, 1989). However, the acceleration is more progressive in the experimental data than in the simulations. This may be related to the estimation of the delay $Die(n)$ which is based only on data from a Pioneer 3377 cultivar (20 leaves). Obviously, the validity for other cultivars, especially with a different number of leaves, must be investigated further.

Number of simultaneously expanding leaves. Only visible leaves, not all expanding leaves, are taken into account in field measurements, thus we had to estimate this variable from the calculation of total number of leaves (visible and hidden in the whorl) by our model. We approximated the hidden fraction of a growing leaf as the part extending from the bottom of a leaf to a distance equal to the length of the preceding sheath. A leaf is thus counted as visible when its length becomes greater than that of the preceding sheath. Similarly, the visible area is calculated as the difference between the total leaf area and the area of the hidden part.

Figure 13 compares data measured for the DEA maize cultivar by Ruget, Bonhomme and Chartier (1996) to simulations with our model. The value of the parameters ($Nt = 16$, $L_M = 79$ cm) are those given in Ruget *et al.* (1996). Although the simulation reproduces the evolution qualitatively, it significantly underestimates the number of simultaneously growing leaves, which confirms the ob-

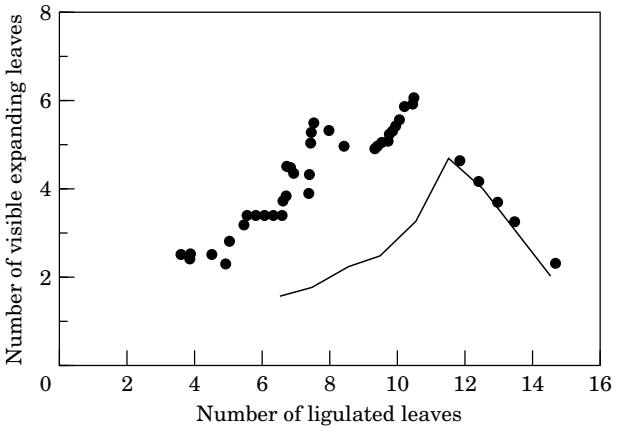


FIG. 13. Measured (symbols) and simulated (line) number of expanding leaves of the whorl, as a function of number of ligulate leaves on main stem, for DEA maize cultivar. Experimental data calculated from Ruget *et al.* (1996). Simulations for $Nt = 16$ and $L_M = 79$ cm.

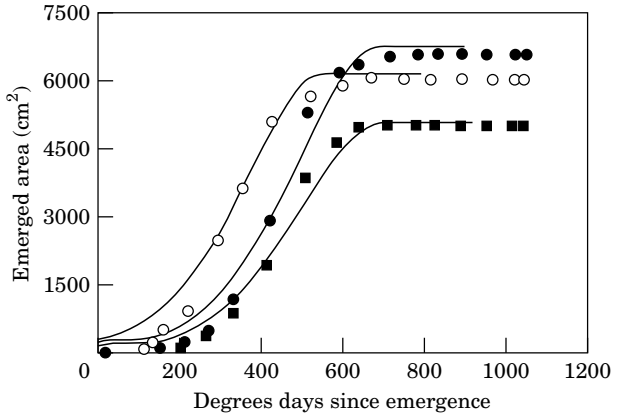


FIG. 14. Emerged leaf area of three maize cultivars, as a function of degree-days since emergence ($T_{base} = 10$ °C). Data (symbols) are from three field experiments by Dwyer and Stewart (1986). Lines represent simulations with Nt and L_M as inputs. Time origin was adjusted so that simulations match experimental data at the time about half the surface was installed.

servation previously made in the comparison with the drawing by Ledent *et al.* (1990). The calculation of the number of simultaneously growing leaves is very dependent on the delay $Die(n)$ so uncertainties in $Die(n)$ could explain the observed bias.

Visible plant area as a function of sum of temperature. Figure 14 compares the measured course of plant leaf area as a function of sum of temperature, to simulations with our model. Experimental data (Dwyer and Steward, 1986) are for two cultivars, grown on different soils during a 2-year field experiment. For simulations, we considered an apex temperature of 20 °C throughout the cycle (i.e. not inducing a reduction in leaf length, see Fig. 4). We calculated Nt and L_M from the final number of leaves and the area of the largest leaf reported by the authors, assuming the leaf shape was as described by eqn (16) and the maximum width was equal to 0·106 L_M (Prévot *et al.*, 1991). The date of emergence

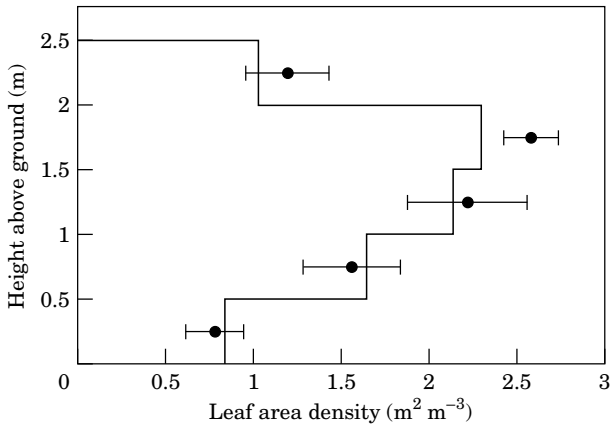


FIG. 15. Leaf area density vertical profile at final vegetative stage. Symbols with error-bars are from experiments by Ivanov *et al.* (1995). Curve results from computation on a simulated cover of maize.

was adjusted so that the curves fit the data at the time half the leaf area was developed. Simulations reproduce the linear part and the end of growth curves well, but somewhat overestimate the leaf area during the beginning of growth. A possible reason is the calculation of degree days from air temperature, which could differ from the apex temperature, depending on the type of soil and the soil water content.

Spatial characterization

Vertical distribution of leaf area density. Vertical distribution of leaf area density is an important pattern governing the distribution of light resources within plant canopies. Figure 15 compares simulated vertical profiles of leaf area density, with field measurements by Ivanov *et al.* (1995), using the 'silhouette' method (Bonhomme and Varlet-Grancher, 1978). Data are from a fully developed, irrigated plot of DEA cultivar, grown in 1993, at usual agronomic density (eight plants m^{-2}). For the simulation, the final number of leaves was set to 16, corresponding to field measurements, and L_M was set to 90 cm so that measured and simulated *lai* were equal. The good agreement between measurements and simulation validates the parameterization for final size of sheath, blade and internode, as well as the geometric representation of leaves for DEA grown under usual agronomic conditions.

Simulation of ground cover as a function of leaf area index. Ground cover (% C) is another important variable, representative of the ability of the crop to intercept sunlight. Field measurements of % C and *lai* were performed in 1989 on an irrigated plot of DEA cultivar. Ground cover was measured using a tele-lens positioned at a height of 10 m and represented a narrow angular range ($\pm 10°$) around the vertical. More details are given in Andrieu, Allirand and Jaggard (1997). The *lai* was measured with great care, using marks on the growing leaves to specify precisely the visible fraction (see Ruget *et al.*, 1996 for details). The actual field row spacing (80 cm) and plant density (nine plants m^{-2}) were introduced in the simulations. Nt was set to 16 and L_M to 80 cm. Simulations were performed for the whole period of leaf area development, and at each time step we estimated

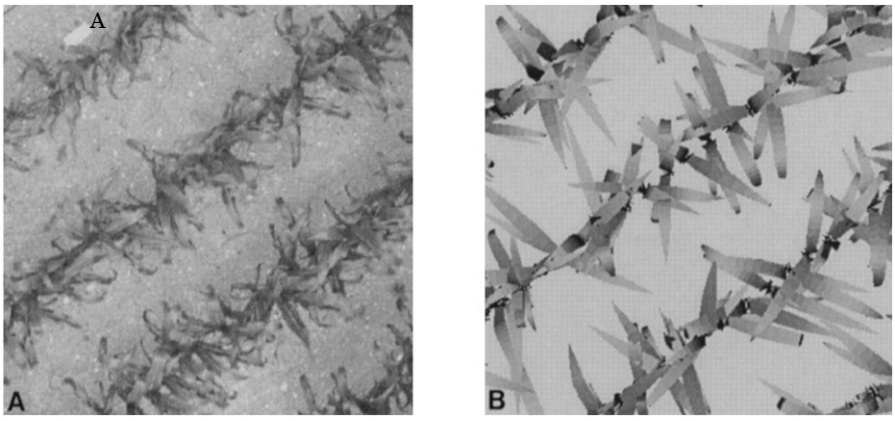


FIG. 16. Real (A) and simulated (B) pictures of a vertical view of three rows of maize.

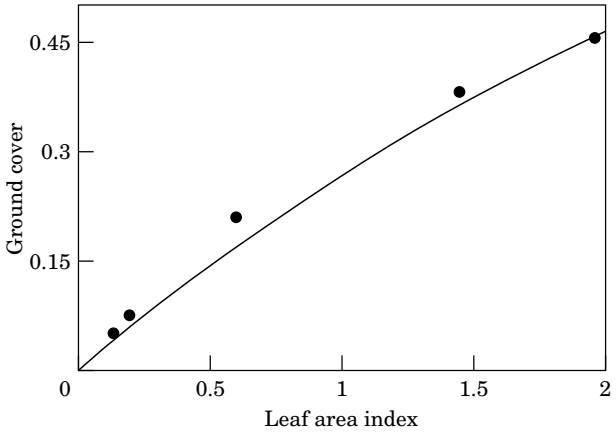


FIG. 17. Ground cover as a function of leaf area index. ●, Data from field experiments on the DEA maize cultivar by Andrieu *et al.* (1997). Simulation with $Nt = 16$ and $L_M = 80$ cm.

the visible leaf area with the same method as described previously. We calculated $\%C$ from the 3D output of the model, using a projection algorithm that reproduces the exact geometric view of the field measurements. Figure 16 shows a photograph of the plot, together with the model output for the same stage of development. The *lai*—$\%C$ relationship simulated by the model (Fig. 17) is very close to the measured one; the extinction coefficient being 0·312 for simulations and 0·324 for measurements. The standard error between measured and simulated $\%C$ is 0·02, within the expected range given the accuracy of *lai* and $\%C$ measurements.

DISCUSSION

This work aimed to evaluate the possibility of developing a process-based 3D architectural model of growth of individual plants within a population. It includes several sets of submodels, most of them in an intermediate stage of development, and we will discuss successively the (1) physiological; (2) geometrical; (3) computing aspects involved.

Physiology

Compared to other crops, modelling maize development may be considered straightforward, due to the simple architecture of the plant. We aimed to define a parameterization that retains the main aspects of vegetative development independent of genotype. However, little information was available at the organ level compared to that addressing global variables such as *lai*.

The parameterization of leaf growth rate and leaf growth duration as a function of physiological time appears to be adapted to a wide range of genotypes. It also seems appropriate to introduce future improvements such as the regulation of leaf elongation and final leaf size by water vapour deficit (Ben Haj Salah and Tardieu, 1996). The parameterization should also be appropriate to account for competition for light which changes leaf width but has a limited effect on rate and duration of elongation (Popa, pers. comm.). On the other hand, the assumption that leaf elongation occurs at a nearly steady rate is a rough approximation in the case of the shortest first and last leaves, and a more detailed description would be required to improve the ability of the model to describe the synchronization between events. Also, the estimation of the physiological duration of leaf elongation should be improved by direct measurements of final leaf length, rather than indirect estimates from leaf area and width/length ratio.

Less information was available for the parameterization of internode elongation. The synchronization between the transition of phase of the apex and the beginning of internode elongation appears stable amongst genotypes, as is the general shape of the profile of internode length as a function of rank. However, the final length and the rate of elongation of internodes are strongly affected by resource availability and by light signals, so that there is no real evidence that physiological time is an appropriate concept on which to base a parameterization of internode elongation. These aspects should be investigated further, to introduce inter-plant competition for light, as well as source-sink relationships in the plant.

Architectural modelling requires knowledge of synchronization between events, such as the start and the end of

module formation. Little is known about these synchronizations, especially about the start of the events, since they take place hidden in the whorl and are difficult to observe. Most data deal with the four or five central phytomers and the extrapolation to first or last phytomers is questionable. The production of primordia by the apex is probably best documented, even if little is known about the effect of environmental factors other than temperature and photoperiod. Conversely, we found no data on the start of leaf elongation. Our calculations show that *Die*, the delay between primordia initiation and leaf elongation, is large and varies with leaf rank, and we found that the response to temperature followed that of the rate of initiation of primordia by the apex rather than that of the rate of leaf elongation. But we lack information on variability between genotypes and the effect of environmental factors. Moreover, there is no evidence of an internal clock scheduling that event: the start of elongation of a leaf could be induced by some other developmental events in the plant. Similarly, synchronization of the start of internode elongation with the end of previous leaf elongation is compatible with only a few published data and is purely correlative. Yet, these synchronisms are important to understand source-sink relationships in the plant; and they could also be a key to understanding the evolution of final organ size as a function of rank. An interesting potential of architectural modelling is to enable simulation of visual observations, such as rate of tip or ligule appearance, which are easy to measure and could be used to test hypotheses of synchronization on a wide range of genotypes or environmental conditions.

Geometry

The 3D representation of the plants makes it possible to simulate visual observation. It also represents the interface between the plants and their environment. We have seen that 3D parameterization, used satisfactorily, represents the main features of surface and gap fraction distribution at the canopy level. However two points at least deserve comments: (1) we gave limited attention to the 3D description of growing leaves in the whorl. Improving this description is required to estimate the trophic conditions of growing leaves, to relate emergence of leaf tips to other developmental events and to improve the possibility of simulating, on virtual plants, the phenological notations usually done on real plants; (2) in our approach, leaf midrib curvature and azimuth are obtained by a statistical approach. This is justified by the lack of knowledge on how these parameters depend on physiological processes, either through mechanical properties of tissue (Moulia, Fournier and Guitard, 1994) or through active regulation such as tropism or obstacle avoidance (Girardin, 1992; Drouet and Moulia, 1996). Incorporating detailed interactions between the 3D shape of organs and physiological processes within the framework of L-systems seems possible *a priori*, but is probably incompatible with the use of homomorphisms, and would greatly increase computational complexity. More information is required on the determinism of these processes and the importance they may have in resource competition between organs or between plants.

Modelling with environmental L-systems

We found that L-systems represent a convenient approach to integrate physiological models from the organ to the plant level. However, we found that the software Graphtal does not enable the transfer of variables between modules, and this represents a significant drawback. We overcame this difficulty by repeating the calculations of variables in the production rules of every module needing them. For instance, the calculation of apex stage of development is done in production rules relative to the apex module, but also in those relative to the internodes, so that an internode 'knows' when the apex converts to the reproductive phase (and thus when it is time for stem elongation). Transfer of information between adjacent modules is possible in contextual L-systems (Prusinkiewicz and Lindenmayer, 1990). Such a feature should prove very useful in avoiding redundant calculations.

Environmental models allow, together with the bi-directional interface, interactions between plants and their environment to be computed at the organ or sub-organ level. Our growth model includes two environmental models: (1) a radiative model computing the distribution of direct sunlight and skylight on the 3D structure. This approach is appropriate for computing the distribution of photosynthetically active radiation and thus the production of dry-matter by individual plants. Extending these calculations to wavelengths where leaf absorption is weak requires multiple light scattering to be taken into account. This would make it possible to introduce e.g. photomorphogenetic response to far-red light. Exact calculation of multiple scattering on a 3D canopy requires large computing resources. However, recent progress has been made to define the proper approximations (Chelle, 1997) and these should result in accurate calculations of radiative fluxes fast enough to be integrated in a growth model; (2) a model for apex temperature. The model that we implemented is based on the energy budget of plant apices, using the 3D description for the radiative budget, and the continuous medium analogy to describe the environment and thus the exchanges of latent and sensible energy. This approach enabled us to validate the principle of bi-directional communication between plants and their environment. The main drawback is the very simple description of the soil and atmospheric environment we introduced in the calculation. Further development should include a more comprehensive 1D soil-vegetation energy transfer model such as the CUPID model (Norman, 1979) which would enable calculation of organ temperature at any stage of development.

CONCLUSION

We presented an approach that models the growth of individual plants within a canopy and implemented it to develop a 3D maize growth model. Clearly, most physiological and physical submodels require further improvement. However, even in this preliminary version, the model enabled a realistic 3D description of maize at any stage of development, with detailed information on the state of the organs in the plant. We believe that this example demon-

strates that environmental L-systems present a previously unmatched capability of integrating processes from the level of organs to that of a plant and of a population of plants competing for resource acquisition. Compared to models using the analogy to a continuous medium, in which individual organs and plants are not identified, we see two domains where this approach presents unique advantages: (1) the widening of the range of virtual experiments to investigate growth process, since essentially any recognized or hypothetical rule can be introduced at the organ level to evaluate its consequence at the plant and canopy level; and (2) the ability to take fully into account the spatial heterogeneity within a field as well as the variability in response of individual plants, both in mono- and pluri-specific populations.

ACKNOWLEDGEMENTS

We thank Yassine Sohbi for implementation of the procedure for bidirectional communication in the Graphtal software and Pierre Cellier for the code of his model. Thanks to J. M. Allirand, R. Bonhomme, M. Chartier and B. Moulia for useful discussions. C. Fournier is the recipient of an Agence de l'Environnement et de la Maîtrise de l'Energie (ADEME) fellowship.

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